

RESEARCH ARTICLE

Robust Classification of UWB NLOS/LOS Using Combined FCE and XGBoost Algorithms

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ABSTRACT Ultra-wideband (UWB) positioning systems are widely applied in localization. However, in complex environments with numerous obstacles, Non-Line-of-Sight (NLOS) propagation of UWB signals can occur, significantly affecting accuracy. Effective identification of Line-of-Sight (LOS) and NLOS signals is essential for precise localization. In this study, we propose a novel NLOS/LOS recognition method based on Extreme Gradient Boosting (XGBoost). The method begins with feature selection using the Pearson Correlation Coefficient to filter less correlated features of the UWB Channel Impulse Response (CIR) data, followed by outlier handling. Preliminary NLOS identification and classification are performed using Fuzzy Comprehensive Evaluation (FCE), with subsequent optimization of FCE weights via the Pattern Search algorithm (PSa). Final classification and recognition are then achieved through XGBoost. This approach, initially trained in one scenario, demonstrates seamless transition and strong generalization across six additional scenarios. Compared to traditional machine learning and neural networks, it offers lower system requirements. Experimental results on an open-source dataset covering seven scenarios show average accuracy, precision, recall, and F1 score of 93.2%, 94.8%, 90.3%, and 92.5%, respectively, highlighting the method's superior performance and robustness in diverse environments.

INDEX TERMS UWB, CIR, FCE, XGBoost algorithm, machine learning, NLOS, indoor positioning.

I. INTRODUCTION

With the rapid development of Internet of Things (IoT) technology, significant progress has been made in mobile communication and localization techniques. Traditional outdoor positioning systems, such as the Global Positioning System (GPS) and the BeiDou Navigation Satellite System (BeiDou), although highly developed, encounter substantial limitations in indoor environments due to signal attenuation and reflection. These issues undermine the reliability of satellite-based positioning technologies, creating a critical need for specialized indoor localization techniques.

To address these challenges, several indoor positioning methods have been developed, including laser positioning, infrared, Wi-Fi, Bluetooth, and Radio Frequency

Identification (RFID). However, these technologies often suffer from radio frequency interference, which limits their accuracy in complex indoor settings [1], [2]. Among these alternatives, Ultra-Wideband (UWB) technology has emerged as a particularly promising solution. UWB, a wireless radio frequency (RF) technology, is valued for its ability to deliver highly accurate ranging and positioning, as well as its capacity to store extensive Channel Impulse Response (CIR) data, facilitated by the DW1000 chip [3]. This capability has made UWB localization systems increasingly significant across various industries, including retail, mining, underwater exploration, and manufacturing [4], and has expanded its application into critical fields such as medical assistance [5], [6] and tourism.

While UWB technology offers considerable advantage, its performance can be compromised in indoor environments, which often contain obstacles like walls, furniture, and human

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bodies [7], [8], [9]. These obstacles lead to Non-Line-of-Sight (NLOS) propagation, introducing signal reflections and multipath effects that result in measurement errors and reduced localization accuracy. NLOS/LOS identification is critical in addressing this challenge, as accurate identification of NLOS conditions plays a vital role in supporting precise location estimation. By distinguishing between NLOS and Line-of-Sight (LOS) conditions, localization systems can apply correction techniques to mitigate the impact of signal interference, thereby improving the overall positioning accuracy. Moreover, beyond location estimation, NLOS/LOS recognition has potential applications in behavioral perception, such as detecting human movement or activity recognition, further broadening its utility in diverse indoor environments.

Various positioning algorithms, such as SPOTTER [10], IDWPSOInLoC [11], and WDMA-UWB [12], have been developed to improve indoor localization and perception tasks. These methods rely on accurate NLOS/LOS identification to mitigate signal interference and enhance localization precision. While our study primarily focuses on NLOS/LOS classification in UWB systems, the outcomes of this research can be integrated into such positioning algorithms to further enhance their performance in complex indoor environments.

To overcome the challenges posed by NLOS conditions, research has increasingly focused on machine learning-based methods for NLOS/LOS recognition. Techniques like Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF) have been employed to process UWB signal's CIR data. More recently, deep learning approaches, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and hybrid CNN-LSTM models, have been introduced. While these methods are effective, their reliance on extensive iterations can hinder classification accuracy [13].

Among the various machine learning algorithms, Extreme Gradient Boosting (XGBoost) has garnered significant attention for its robust classification capabilities. Introduced by Chen and Guestrin [14], XGBoost is a Gradient Boosting Decision Tree (GBDT) algorithm known for its efficiency in distributed computing and exceptional model performance. Its widespread application across domains, such as healthcare [15] and drug discovery [16], is largely due to its high problem-solving efficiency and reduced need for extensive feature engineering. Moreover, XGBoost is particularly well-suited to scenarios with smaller datasets, making it an ideal choice for studies like ours, where sample sizes are limited.

Building on the strengths of XGBoost, Jia et al. [17] proposed a novel NLOS signal identification method that leverages the algorithm's powerful classification capabilities. This method integrates multiple input features into the XGBoost model, significantly enhancing the accuracy of NLOS signal identification. However, the study also highlights a notable challenge: while XGBoost proves effective in many scenarios, its accuracy can decrease in complex indoor environments, with performance dropping to approximately 86.2%.

While XGBoost has demonstrated strong performance in recognition and classification tasks, its application in NLOS identification within UWB localization is relatively underexplored. The existing literature on XGBoost in this domain is limited, particularly concerning its performance in complex environments.

In this study, we assess the effectiveness of the XGBoost model in NLOS identification across seven distinct environments, highlighting its capabilities in error mitigation and enhancing localization accuracy. The main contributions of this work are as follows:

- 1) We processed the CIR features using Pearson Correlation Coefficient (PCC) analysis, selecting six features with lower correlations to ensure stronger feature representation and improved robustness in classification performance.
- 2) A two-stage classification approach was employed for UWB CIR feature processing. In the first stage, outlier handling was performed using a combination of Interquartile Range (IQR), Z-score, and DBSCAN (Density-Based Spatial Clustering of Applications with Noise) methods to effectively identify and mitigate data anomalies. After the outliers were removed, preliminary NLOS identification was conducted using Fuzzy Comprehensive Evaluation (FCE). The results of the FCE process were then further optimized through the Pattern Search Algorithm (PSa), ensuring that the feature weights were adaptively adjusted to improve classification performance. In the second stage, the optimized features were fed into the XGBoost model for final NLOS/LOS classification, leveraging its robust classification capabilities to achieve high accuracy.
- 3) We propose a novel method combining FCE and XGBoost for NLOS identification and classification. This approach was rigorously tested across six additional environments and compared against existing techniques, including DT, CNN-LSTM, Morlet Wavelet Transform-CNN (MWT-CNN), and Fully Convolutional Networks with Attention (FCN-Attention). Our FCE-XGBoost method demonstrated superior accuracy, precision, recall, and F1 scores, showcasing its robustness and effectiveness under complex NLOS conditions.

The remainder of this article is structured as follows. Section II provides a concise review of related work on UWB NLOS and LOS recognition. Section III details the methodology, with a focus on the FCE and XGBoost models. Section IV offers an in-depth analysis of the simulation results across seven different environments. Finally, Section V summarizes the paper by highlighting the key findings and contributions of the study.

II. RELATED WORK

The rapid development of UWB technology has revolutionized indoor positioning systems, making it possible to perform precise positioning in environments where conventional systems fail. However, one of the most significant

challenges that remains is the identification of NLOS conditions, which can severely degrade positioning accuracy. Effective identification of NLOS errors is essential for enhancing the reliability of UWB-based systems. Over the past few years, extensive research has been devoted to developing innovative methods for NLOS identification. However, challenges in accuracy and robustness remain in different indoor environments [18].

One of the earliest approaches to NLOS identification is based on statistical methods, which distinguish LOS and NLOS signals by analyzing the statistical characteristics of ranging errors. For example, Momtaz et al. [19] proposed a method that utilizes the statistical properties of ranging errors for NLOS identification. The primary advantage of this method lies in its simplicity; however, it is highly sensitive to environmental changes and requires substantial data support. Building on this, Guo et al. [20] introduced the Residual Hypothesis Test Algorithm (RHTA) for hyperbolic positioning in mixed LOS-NLOS environments. This method differentiates between LOS and NLOS base stations by calculating the residuals between initial and refined position estimates, without prior knowledge of NLOS measurements. Although RHTA effectively reduces positioning errors caused by NLOS, its performance diminishes under high noise conditions.

Another approach focuses on NLOS identification using CIR feature parameters, particularly through machine learning (ML) and deep learning (DL) techniques. Marano et al. [21] proposed a method utilizing CIR feature extraction combined with SVM for NLOS identification, which mitigates ranging errors in NLOS conditions. This method eliminates the need for statistical modeling but is computationally intensive. Wu et al. [22] expanded on this by extracting features such as energy, peak amplitude, rise time, mean excess delay, and kurtosis from CIR data, employing SVM for NLOS identification. While this approach improves positioning accuracy, it demands extensive training data and significant hardware resources. Park et al. [23] addressed data imbalance issues in UWB LOS/NLOS identification models through transfer learning (TL), enhancing generalization across different environments. Despite its complexity and reliance on pre-trained models and substantial annotated data, this approach has proven effective. Yang et al. [24] also applied SVM to classify CIR feature points for UWB sensor-based indoor LOS/NLOS localization, achieving higher classification accuracy, though the method remains sensitive to noise and outliers. These studies demonstrate that ML techniques, through CIR feature extraction, classification, and TL, can significantly enhance NLOS identification accuracy and robustness. However, challenges related to computational resource demands and environmental adaptability persist.

DL techniques have also been explored for NLOS identification. Shirin Abadi and Abbasi [25] introduced CNN for UWB channel classification, which significantly improved classification accuracy but required high computational resources. Jiang et al. [26] combined CNN and LSTM

networks to automatically extract CIR features and classify LOS/NLOS conditions, proposing a novel classification method. Although this approach enhances classification performance, it involves high model complexity and demands extensive training data. In another study, He [27] proposed a classification method based on denoising UWB CIR data, which improved classification accuracy and robustness, though it might lead to the loss of useful signal features. Cui et al. [28] integrated MWT and CNN to extract CIR features for UWB LOS/NLOS identification, significantly enhancing accuracy by combining time and frequency domain features. However, the generalization of this model in dynamic scenarios remains limited. Pei et al. [29] proposed an NLOS identification method based on FCN and attention mechanisms, which improved model precision and robustness but required high complexity and lengthy training and inference processes. Deng et al. [30] refined the CNN structure for NLOS identification using CIR feature parameters, achieving classification through an improved CNN. While effective, this approach demands considerable hardware resources, rendering it unsuitable for resource-constrained devices. These studies illustrate that DL techniques can significantly enhance NLOS identification performance through automatic feature extraction and advanced network structures. Nonetheless, challenges related to computational resources and environmental adaptability remain.

III. METHODOLOGY

A. SYSTEM ARCHITECTURE OVERVIEW

The proposed method, as illustrated in Fig. 1, provides a comprehensive overview of the method design and data flow. This framework integrates the FCE and XGBoost algorithms for efficient NLOS identification and classification. Initially, the CIR dataset undergoes several preprocessing steps, including PCC analysis, data selection, outlier handling, and normalization. The FCE algorithm computes membership degrees, optimizes weights, determines thresholds, and evaluates F1 scores using PSa. Simultaneously, the XGBoost algorithm processes the data, optimizes hyperparameters, trains the model, and assesses its accuracy. The best-performing XGBoost model is then selected for making predictions. The integrated approach is evaluated and demonstrates enhanced accuracy and robustness in practical NLOS identification and classification scenarios.

B. CIR FEATURE SELECTION

To develop our NLOS identification model, according to [31], we extracted several features from the CIR signal waveform. These features serve as candidate classification parameters that are critical for distinguishing between LOS and NLOS conditions. The selected features include Distance Measurement (RANGE), First Path Index (FP_IDX), First Path Amplitude 1 (FP_AMP1), First Path Amplitude 2 (FP_AMP2), First Path Amplitude 3 (FP_AMP3),

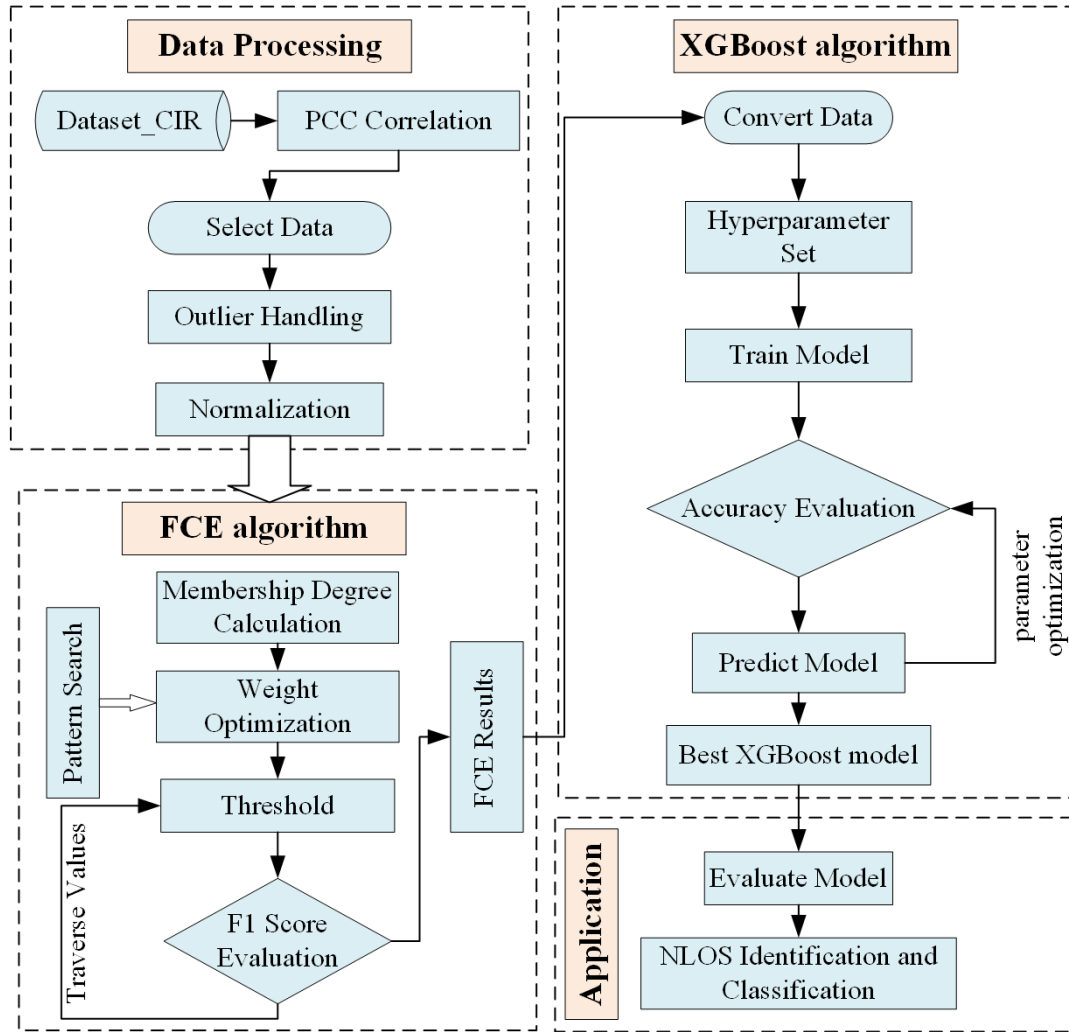


FIGURE 1. Flowchart of FCE-XGboost.

Standard Deviation of Noise (STDEV_NOISE), Channel Impulse Response Power (CIR_PWR), Maximum Noise Level (MAX_NOISE), and Received Power Accumulated (RXPACC).

The extraction of these features is based on specific considerations related to signal behavior in different propagation environments. For instance, when a signal penetrates obstacles, its energy significantly attenuates, which can be observed through changes in the amplitude features such as FP_AMP1, FP_AMP2, and FP_AMP3. Additionally, noise levels, represented by STDEV_NOISE, may vary considerably across different propagation channels, especially when the channel is affected by multiple obstructions.

Furthermore, RANGE and CIR_PWR are particularly important in scenarios where the received signal includes multipath components. These features tend to increase as the presence of multipath effects becomes more pronounced. In contrast, in a LOS scenario where multipath effects are minimal or absent, the RXPACC is typically higher. However,

when multipath is present, features such as MAX_NOISE and STDEV_NOISE show significant increases, reflecting the added complexity in the signal environment.

The calculation methods for these candidate features are essential for accurately capturing variations in signal characteristics under different propagation conditions. By analyzing how each feature responds to changes in the environment, we can better understand the underlying factors that influence NLOS identification, thereby improving the robustness and accuracy of our model. To this end, we extracted and analyzed specific CIR features, each of which plays a crucial role in enhancing our model's performance. In the following, we will outline the calculation methods and the importance of each feature in the NLOS identification process.

(i) RANGE

$$\hat{d} = c \times \text{ToA} = \begin{cases} d + e_{\text{LOS}}, & \text{if LOS} \\ d + e_{\text{NLOS}} + b, & \text{if NLOS} \end{cases} \quad (1)$$

The measurement range is calculated through the Time of Flight (ToF), where $c = 3 \times 10^8$ m/s represents the speed of light, d denotes the actual distance between the transmitter and receiver, e_{LOS} accounts for the ranging error in LOS scenarios, e_{NLOS} represents the ranging error in NLOS scenarios, and b is the bias introduced by multipath propagation.

- (ii) **FP_IDX**: It utilizes the high temporal resolution of UWB signals to determine the ToA by detecting the first significant pulse in the received CIR. It reflects the combined effects of multipath propagation. The first significant pulse in the CIR sequence is usually detected by setting a threshold value, and its corresponding index is called **FP_IDX**. It helps in determining the ToA and distinguishing between LOS and NLOS scenarios.
- (iii) **FP_AMP1**, **FP_AMP2**, and **FP_AMP3**: The amplitudes of different parts of the first path, obtained from the DW1000 diagnostic registers, represent the strength of the signal. These amplitude values are critical for understanding signal quality, as they reflect the signal's intensity during transmission. In this context, the maximum amplitude of the received signal, r_{max} , is determined by the equation:

$$r_{\text{max}} = \max \{|r(t_i)|\}, \quad (2)$$

where r_{max} signifies the peak value of the absolute amplitude of the received signal $r(t_i)$ over time. Here, $|r(t_i)|$ represents the absolute value of the received signal at a specific time t_i . Identifying this peak amplitude is essential for accurately assessing the signal's strength and ensuring reliable performance in various signal transmission scenarios.

- (iv) **STDEV_NOISE**: This metric is used to quantify the standard deviation of noise present in the signal. It is calculated as:

$$\sigma_n = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (3)$$

where x_i are the noise samples, $\bar{x}$ is the mean of the noise samples, and N is the number of samples.

- (v) **CIR_PWRCIR**

$$P_{\text{CIR}} = 10 \times \log_{10} \left(\frac{G_{\text{CIR}} \times 2^{17}}{P_{\text{RX}}^2} \right) - A \quad (4)$$

The total power of the entire CIR is represented by this formula, where G_{CIR} denotes the maximum growth of the CIR power, P_{RX} indicates the count of received preamble symbols, and A is a predefined constant of the system.

- (vi) **MAX_NOISE**

$$\delta_{\text{snr}} = 10 \log_{10} \left(\frac{r_{\text{max}}^2}{2\sigma_n^2} \right) \quad (5)$$

where δ_{snr} represents the signal-to-noise ratio, r_{max} is the maximum amplitude of the signal, σ_n is the standard deviation of the noise. It can indicate the relative strength of the signal in a noisy environment.

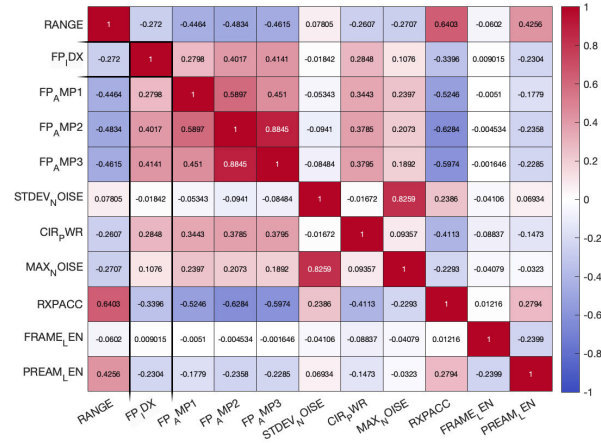


FIGURE 2. Heatmap of Pearson correlation coefficients for CIR features.

The CIR features we have selected serve as the foundation for our NLOS identification model. To further refine this model, we perform a PCC analysis to quantify the correlations between these features, thereby improving the model's performance and reliability. The following outlines the steps for calculating the PCC to identify the correlations between features [32].

$$\text{PCC} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}} \quad (6)$$

where:

- x_i and y_i are the i -th samples of the two variables,
- $\bar{x}$ and $\bar{y}$ are the sample means of the two variables,
- N is the number of samples.

Furthermore, the PCC is calculated by combining the covariance and the standard deviations into the PCC formula (6). The correlations between features are visualized in Fig. 2 using a heatmap. Based on the analysis results, the six features with the lowest correlations were selected for subsequent modeling. These features are: RANGE, STDEV_NOISE, CIR_PWR, MAX_NOISE, RXPACC, and FP_AMP3. The low correlation among these features helps to reduce multicollinearity issues, thereby enhancing the robustness and accuracy of the model.

C. NLOS IDENTIFICATION AND HANDLING OUTLIERS

Before performing NLOS identification, removing outliers is a critical step. In UWB indoor positioning systems, outliers typically arise due to multipath effects, signal reflections, interference, and noise. These outliers can adversely affect positioning accuracy and the robustness of the model. The benefit of removing outliers lies in improving data quality, reducing errors, and thereby enhancing the model's accuracy.

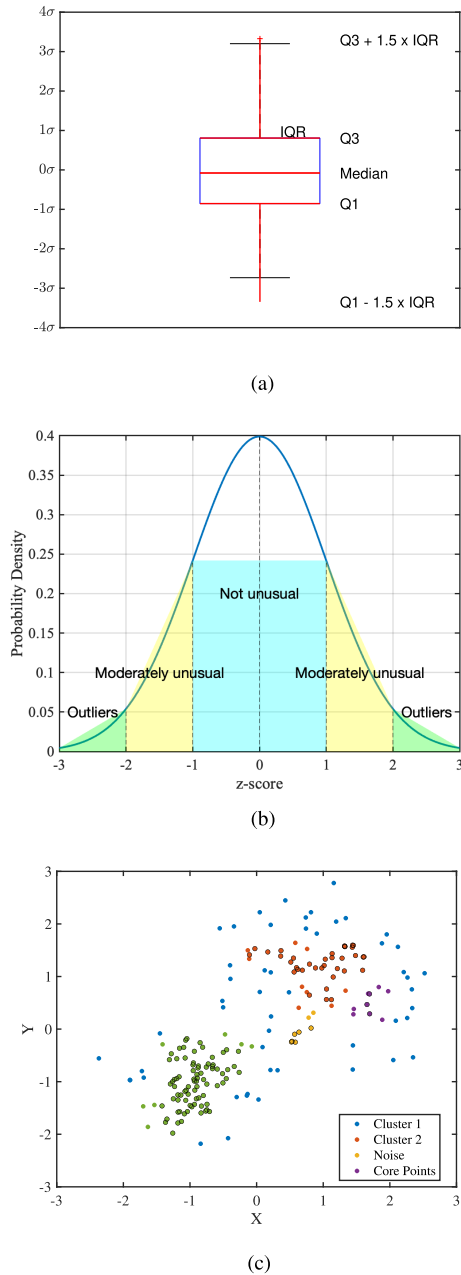


FIGURE 3. Comparison of different outlier detection methods.(a) Box plot with IQR,(b) DBSCAN clustering,(c) Detecting outliers with Z-Scores.

and stability. We employ three methods to remove outliers: IQR, Z-score, and DBSCAN. The following provides a detailed description of these three methods. They are illustrated in Fig. 3.

The IQR is a widely used method for detecting outliers in data sets [33]. The fundamental principle of IQR-based outlier detection is to utilize the quartiles of the data to identify anomalies. Specifically, data points that fall outside the range defined by the lower bound (LB) and upper bound (UB) are considered outliers.

The calculation process begins with determining the first quartile ($Q1$) and the third quartile ($Q3$) of the dataset. The interquartile range is then calculated using the formula $IQR = Q3 - Q1$, which represents the spread of the middle 50% of the data.

Once the IQR is calculated, the next step is to identify outliers. Any data point that is smaller than the LB or larger than the UB is classified as an outlier. This method effectively isolates data points that deviate significantly from the central tendency, as defined by the quartiles. The formula used to determine outliers is as follows:

$$\begin{aligned} IQR &= Q3 - Q1 \\ LB &= Q1 - 1.5 \times IQR \\ UB &= Q3 + 1.5 \times IQR \end{aligned} \quad (7)$$

The Z-score method is a statistical approach for identifying outliers by measuring how far each data point deviates from the mean of the dataset [34]. The underlying principle is that data points with an absolute Z-score greater than a certain threshold (typically 3) are considered outliers.

To apply this method, the mean μ and standard deviation σ of the dataset are first calculated. Next, the Z-score for each data point is computed, indicating how many standard deviations the data point is from the mean. Finally, any data point with an absolute Z-score greater than 3 is classified as an outlier.

$$Z = \frac{x - \mu}{\sigma} \quad (8)$$

where x is a data point, μ is the mean of the data, and σ is the standard deviation of the data.

DBSCAN is a widely used algorithm for identifying outliers within data sets [35]. The core principle of DBSCAN is to detect dense regions in the data, classifying points that are isolated or found in sparse regions as outliers.

The implementation of DBSCAN follows a systematic process. Firstly, appropriate parameters for the radius ϵ and the minimum number of samples $MinPts$ must be selected. In this case, after several rounds of parameter tuning and experimentation, ϵ is set to 0.5 and $MinPts$ to 5. These values were chosen based on the specific distribution and characteristics of our dataset, as they provided a good balance between detecting meaningful clusters and identifying outliers effectively. For each data point, the algorithm assesses whether the number of neighboring points within its ϵ -neighborhood meets or exceeds $MinPts$. If this criterion is fulfilled, the point is designated as a core point; otherwise, it is classified as a border point or a noise point.

Then, the algorithm groups all points within the ϵ -neighborhoods of core points into distinct clusters. Points that do not associate with any cluster are identified as outliers.

Specifically, the ϵ -neighborhood of a point p is defined as:

$$N_{\epsilon}(p) = \{q \mid \text{distance}(p, q) \leq \epsilon\}, \quad |N_{\epsilon}(p)| \geq MinPts \quad (9)$$

In our case, where $\epsilon = 0.5$ and $MinPts = 5$, DBSCAN effectively clusters points based on density, detecting and

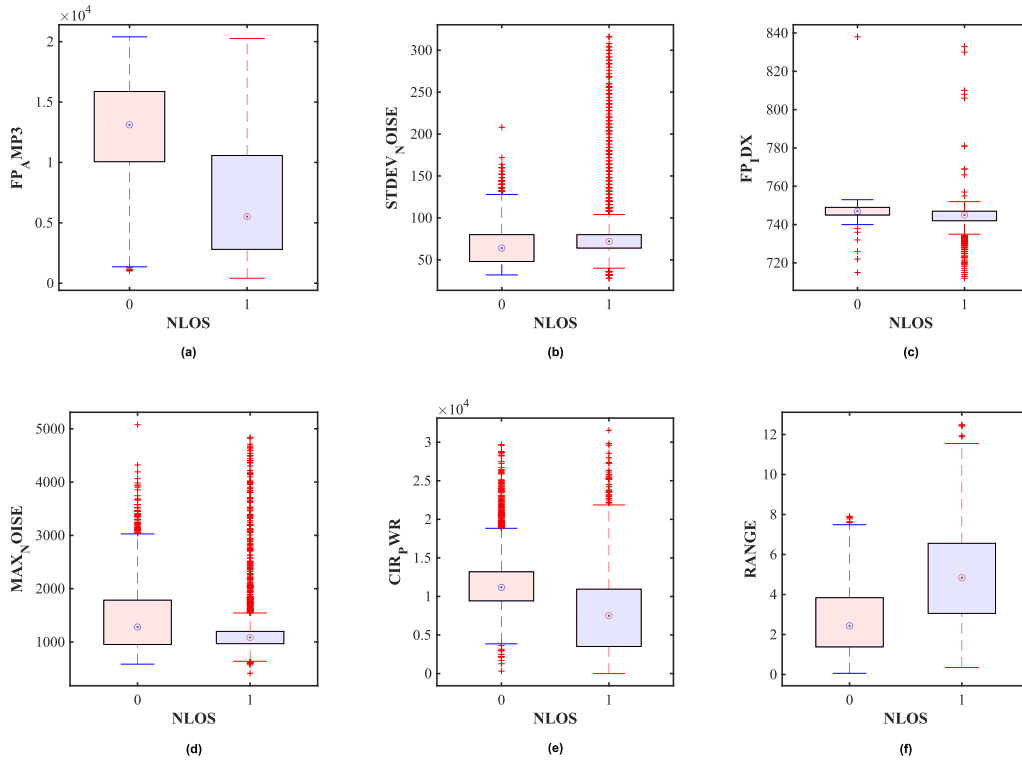


FIGURE 4. Comparison of six features before outlier treatment. (a) FP_{AMP3} , (b) $STDEV_{NOISE}$, (c) CIR_{PWR} , (d) MAX_{NOISE} , (e) CIR_{PWR} , (f) $RANGE$.

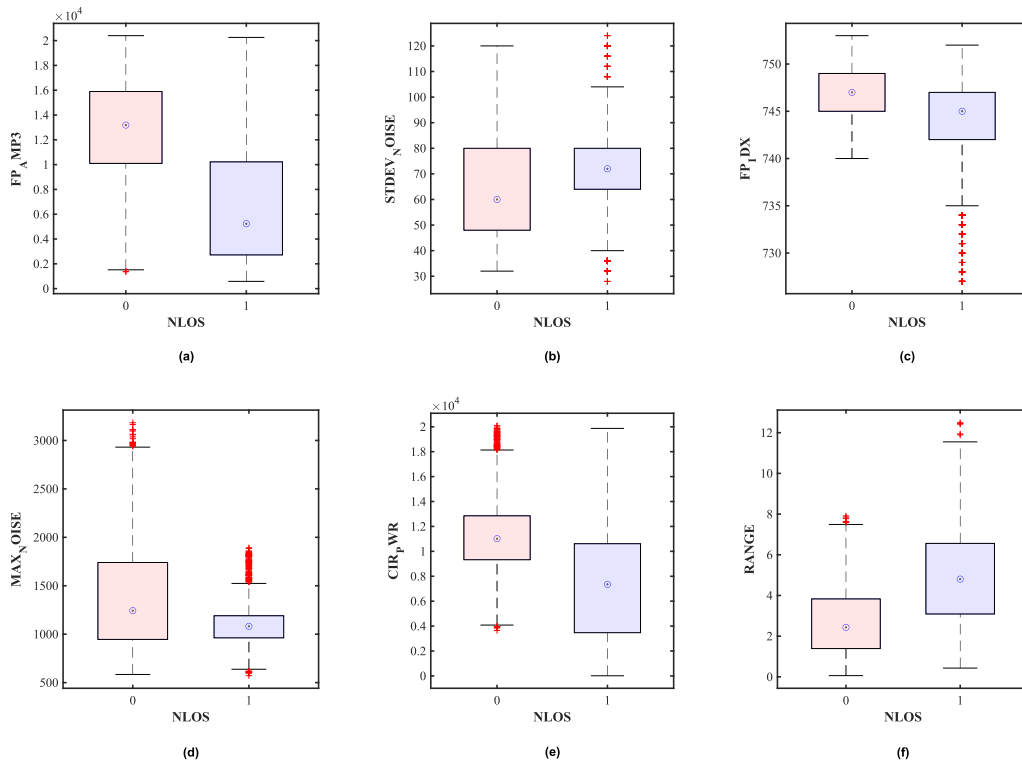


FIGURE 5. Comparison of six features after outlier treatment. (a) FP_{AMP3} , (b) $STDEV_{NOISE}$, (c) CIR_{PWR} , (d) MAX_{NOISE} , (e) CIR_{PWR} , (f) $RANGE$.

labeling sparse or isolated points as outliers. These parameters were carefully tuned through experimentation to ensure

that the algorithm correctly identifies clusters and isolates outliers, given the nature of the data.

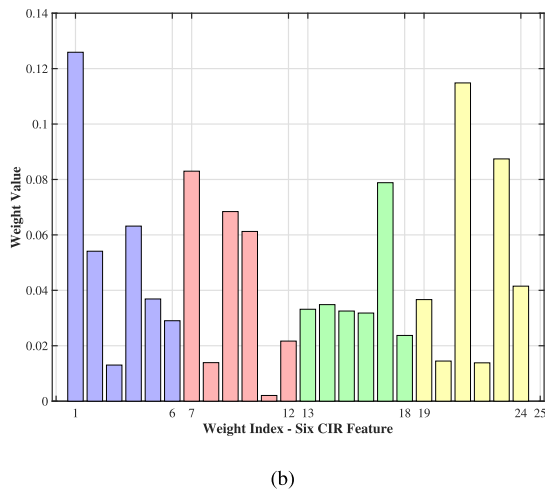
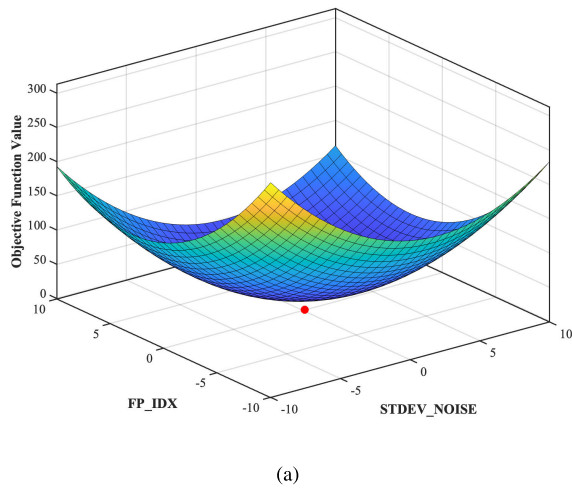


FIGURE 6. (a) Optimal weights found by pattern search, (b) Distribution of optimized feature weights.

As illustrated in Fig. 3(c), DBSCAN successfully identified two main clusters (Cluster 1 and Cluster 2), each formed by densely packed points. The algorithm also identified noise points, which are isolated points that do not belong to any cluster. Core points, which have a sufficient number of neighbors (greater than or equal to $MinPts$), are clearly marked in the figure. Points in the sparse regions that did not meet this criterion were labeled as noise. This visualization confirms the effectiveness of DBSCAN in distinguishing between densely populated clusters and outliers based on the chosen parameters.

The comparison between the data charts before and after processing (Fig. 4 for raw data and Fig. 5 for processed data) demonstrates the significant effects of outlier handling. Initially, the FP_AMP3 and CIR_PWR features, processed using the IRQ method, exhibit obvious outliers in the raw data, marked by red crosses. These outliers are effectively removed post-processing, resulting in a tighter data distribution and reducing the impact of extreme values.

For the STDEV_NOISE, FP_IDX, and MAX_NOISE features, the application of the Z-score method plays a crucial role in reducing outliers and consolidating data distribution. This method successfully eliminates extreme high and low values, leading to a more normally distributed dataset. Similarly, the RANGE features benefit from the outlier identification and removal using the DBSCAN density clustering algorithm. Post-processing, these features show a more compact distribution, minimizing noise points in the data.

Specifically, the FP_AMP3 feature's box plot of the raw data reveals multiple outliers that are distant from the main data cluster. After processing, these outliers are nearly eliminated, concentrating the data within a narrower range. The box and whisker plots demonstrate a clear trend towards centralization, indicating the effectiveness of the outlier handling techniques. In the case of the STDEV_NOISE feature, the raw data contains numerous outliers, which are significantly reduced after Z-score processing. This adjustment results in a more normal distribution, diminishing the impact of extreme values.

In contrast, the MAX_NOISE and CIR_PWR features exhibit highly skewed data distributions due to the presence of outliers. Post-processing with the Z-score method leads to a more symmetrical distribution, reducing the influence of extreme values.

The RANGE feature also shows a significant reduction in noise points after applying the DBSCAN clustering algorithm, which results in data being concentrated within a more reasonable range. This enhances data consistency and reliability. These processing methods not only reduce the interference of extreme values in the analysis results but also ensure that the data distribution aligns more closely with statistical assumptions.

In terms of data outlier processing, ablation experiments were performed to compare the impact of various outlier removal methods on FCE performance using quantitative metrics such as accuracy, precision, recall, and F1 score, as shown in Table 1. The combined use of IRQ, Z-Score, and DBSCAN for outlier removal yielded the highest overall improvement, with accuracy increasing from 0.755 to 0.852 and F1 score from 0.749 to 0.829. This highlights the effectiveness of integrating multiple outlier removal techniques, as each method addresses different types of data irregularities.

The IRQ method excels in recall, reducing false negatives (recall: 0.8421), but at the cost of lower precision. The Z-Score method balances precision (0.7534) and recall (0.7789), making it effective for data following a Gaussian distribution. DBSCAN, with the highest precision (0.7993), efficiently detects spatially distributed outliers. By combining these three approaches, the model benefits from enhanced precision, recall, and overall robustness, as each method compensates for the limitations of the others.

This synergy demonstrates that utilizing multiple outlier detection methods leads to more comprehensive data cleaning

TABLE 1. Comparison of FCE evaluation metrics with and without different outlier removal methods.

Metric	Without Outlier Removal (%)	With Outlier Removal (%)	Only IRQ (%)	Only Z-Score (%)	Only DBSCAN (%)
Accuracy	0.755	0.852	0.6638	0.7651	0.7683
Precision	0.749	0.844	0.6249	0.7534	0.7993
Recall	0.750	0.804	0.8421	0.7789	0.7228
F1 Score	0.749	0.829	0.7174	0.7660	0.7591
Best Threshold	0.465	0.467	0.3807	0.4383	0.4422

Note: These evaluation metrics will be further explained in Section IV-A.

and significantly boosts model performance across all metrics.

D. FCE MODEL

The UWB CIR features' data were collected from experiments or measurements, denoted as $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$, where x_i represents the i -th sample feature value, and y_i represents the corresponding actual value.

To ensure comparability among different features, the collected data must be standardized. The standardization formula is given as:

$$x'_i = \frac{x_i - \mu}{\sigma} \quad (10)$$

where x_i is the original feature value, μ is the mean of the feature, and σ is the standard deviation of the feature. The standardized feature value x'_i has a mean of 0 and a standard deviation of 1.

The FCE method is an integrated evaluation approach grounded in fuzzy mathematics [36]. This method leverages the theory of membership degrees to transform qualitative evaluations into quantitative assessments, making it particularly useful for comprehensive evaluations of objects or subjects influenced by multiple factors [37].

The FCE model, as applied in this context, can be mathematically expressed as follows [38], [39]:

$$\tilde{B} = \tilde{A} \cdot \tilde{R} \quad (11)$$

where $\tilde{A}$ and $\tilde{B}$ are two fuzzy sets defined in the factor domain and evaluation domain, respectively. In this study, $\tilde{A}$ represents the six CIR features, while $\tilde{R}$ denotes the relationship between $\tilde{A}$ and $\tilde{B}$. The operator “ $\cdot$ ” is a fuzzy operator that facilitates the derivation of evaluation results by processing the information contained within $\tilde{A}$ and $\tilde{R}$.

The membership degree of each evaluation factor A_i for each evaluation grade R is determined, resulting in the formation of the membership degree matrix, also known as the Fuzzy Relation Matrix.

To allow the FCE weights to vary and to find the weights that maximize the FCE score, a PSa is employed. The objective function is defined to evaluate the FCE score while including regularization terms to prevent overfitting:

$$J(\theta) = -E_f(\mu_t, y_t, w) + \lambda_1 \sum w^2 + \lambda_2 \sum |w| \quad (12)$$

where E_f is the fuzzy evaluation score function, μ_t denotes the membership degrees of the training data, y_t represents the training labels, and w is the weight vector. The regularization parameters λ_1 and λ_2 control the strengths of L2 and L1 regularization. Algorithm 1 presents the pseudocode for optimizing the weights and threshold in the FCE process.

During the PSa optimization process, the weights for all six features are optimized. To visualize the process, the weights of two representative features, STDEV_NOISE and FP_IDX, are selected.

The optimal solution identified by the PSa algorithm is shown in Fig. 6(a), where the x-axis and y-axis represent the weights of STDEV_NOISE and FP_IDX, respectively. The surface plot visualizes the variation in the objective function across different weight combinations, clearly illustrating the optimization process. The red dot marks the optimal solution, demonstrating the PSa algorithm's effectiveness in minimizing the objective function and identifying the best weight configuration.

The distribution of the optimized weights across all six features is displayed in Fig. 6(b). The bar chart reveals how the algorithm allocates the weights, with higher values indicating a greater influence on the objective function. This visualization highlights the relative importance of each feature in the final FCE calculation.

The FCE evaluation plays a crucial role in the preliminary identification of NLOS conditions by assigning membership degrees to different features. By setting a threshold, the evaluation results are divided into NLOS and LOS categories. The threshold optimization process involves traversing a defined range (e.g., 0 to 1 with small steps), calculating accuracy for each threshold, and selecting the one that yields the highest classification accuracy.

E. XGBoost ALGORITHM AND MODEL ESTABLISHMENT

1) XGBoost ALGORITHM

Feature processing and FCE score calculation are crucial steps in the workflow. It is essential to compute the FCE scores for these CIR features. Assume the FCE scores are: $\{s_1, s_2, \dots, s_n\}$, it uses FCE scores as features into XGBoost mode, the basic principle of XGBoost is to iteratively train multiple weak learners (i.e., decision trees) and aggregate their prediction results through weighted summation to form a strong learner. The following describes

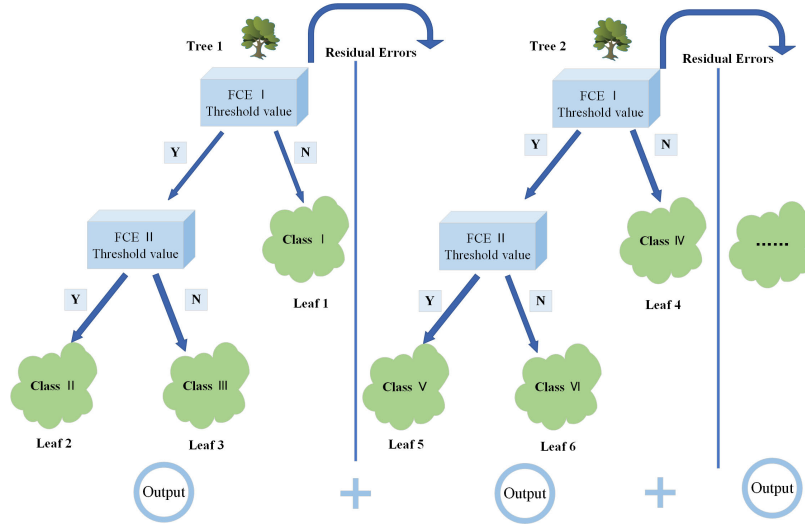


FIGURE 7. Schematic of XGBoost trees.

the specific implementation process. The mechanism of XGBoost involves continuously adding and training new trees to fit the residuals from the previous iteration, as illustrated in Fig. 7.

Specifically, in each iteration, XGBoost generates a new tree aimed at correcting the errors in the previous predictions. By doing so, the model incrementally improves its predictive accuracy. The final prediction for each instance is obtained by summing the scores of the corresponding leaf nodes from all the trees. This approach ensures that the overall predictive performance of the model increases with the number of iterations. The prediction formula of the XGBoost model is as follows [40].

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i), \quad f_k \in \mathcal{F} \quad (13)$$

where $\mathcal{F}$ is the space of all regression trees, and each f_k represents the structure and leaf weights of a tree, $\hat{y}_i^{(t-1)}$ is the prediction result from the previous $(t-1)$ trees, $f_t(x_i)$ is the model of the t -th tree. Here, x_i is the input feature vector, i.e., the FCE scores $\{s_1, s_2, \dots, s_n\}$. XGBoost trains the model by minimizing the following regularized loss function, the loss function is written as:

$$L(\phi) = \sum_{i=1}^n l(\hat{y}_i, y_i) + \sum_{k=1}^K \Omega(f_k) \quad (14)$$

where n is the number of NLOS Features samples, l is the loss function (such as logistic loss), which measures the error between the model prediction $\hat{y}_i$ and the true value y_i . The specific form is:

$$l(\hat{y}_i, y_i) = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (15)$$

The regularization term $\Omega(f_k)$ is represents the sum of the complexities of all t trees, added to the objective function as

a regularization term to prevent the model from overfitting.

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (16)$$

At the t -th iteration, add a new function f_t to minimize the following objective, Gradient Boosting is as follows:

$$L^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (17)$$

To quickly optimize this objective, a second-order Taylor expansion can be used:

$$L^{(t)} \approx \sum_{i=1}^n [l(y_i, \hat{y}_i^{(t-1)}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (18)$$

where g_i represents the first-order derivative of the loss function l with respect to the previous round's prediction $\hat{y}_i^{(t-1)}$, it measures the discrepancy between the current prediction and the true value. and h_i represents the second-order derivative of the loss function l with respect to the previous round's prediction $\hat{y}_i^{(t-1)}$, it measures the curvature of the loss function, helping the optimizer converge more quickly.

$$g_i = \frac{\partial l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}, \quad h_i = \frac{\partial^2 l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)^2}} \quad (19)$$

The optimal weights of leaf nodes Calculates, for a fixed tree structure $q(x)$, the optimal weight w_j^* of the leaf nodes can be calculated as follows:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \quad (20)$$

Algorithm 1 FCE Weight and Threshold Optimization

Input: *membership_degree_train, membership_degree_test, train_labels, test_labels, λ_1, λ_2*

Output: *best_fce_threshold, best_fce_f1*

```

1: Step 1: Define Objective Function for Weight Optimization
2:   Objective function  $J(w) = -E_f(\text{membership\_degree\_train}, \text{train\_labels}, w) + \lambda_1 \sum w^2 + \lambda_2 \sum |w|$ 
3: Step 2: Initialize Optimization Variables
4:   Initialize  $nvars = \text{size}(\text{membership\_degree\_train}, 2)$ ,  $lb = 0$ ,  $ub = 1$ , and  $\text{initial\_guess}$ 
5:    $\text{initial\_guess}$  is a random vector between  $lb$  and  $ub$ 
6: Step 3: Weight Optimization using PSA
7: for each iteration do
8:   Generate candidate solutions  $P_k = \{x_k + \Delta_k d_i \mid d_i \in D\}$ 
9:   Evaluate objective function  $f(p)$  for each candidate  $p$ 
10:  Update  $x_{k+1} = \arg \min_{p \in P_k} f(p)$ 
11:  if no better solution found then
12:    Reduce step size  $\Delta_k$ 
13:  end if
14: end for
15: Save optimized weights  $w^* = x_{k+1}$ 
16: Step 4: Compute FCE Scores with Optimized Weights
17:   Compute  $\text{fuzzy\_scores\_train} = \text{membership\_degree\_train} \times w^*$ 
18:   Compute  $\text{fuzzy\_scores\_test} = \text{membership\_degree\_test} \times w^*$ 
19: Step 5: Threshold Optimization
20: Initialize  $\text{best\_fce\_f1} = 0$ ,  $\text{best\_fce\_threshold} = 0$ 
21: for each  $t$  in  $\text{thresholds} = [0 : 0.0001 : 1]$  do
22:   Calculate  $\text{predicted\_labels\_fce} = (\text{fuzzy\_scores\_test} > t)$ 
23:   Compute classification metrics:  $tp, fp, fn$ , and  $tn$ 
24:   if  $f1\_score$  improves then
25:     Update  $\text{best\_fce\_f1}$  and  $\text{best\_fce\_threshold}$ 
26:   end if
27: end for
28: return  $\text{best\_fce\_threshold}, \text{best\_fce\_f1}$ 

```

where w_j^* represents the optimal weight of the j -th leaf node, I_j represents the set of instances in the j -th leaf node. h_i is the second derivative of the loss function with respect to the prediction, λ is the regularization parameter used to prevent overfitting, and the corresponding optimal value is.

$$\tilde{L}^{(t)}(q) = -\frac{1}{2} \sum_{j=1}^T \frac{\left(\sum_{i \in I_j} g_i \right)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T \quad (21)$$

To evaluate split candidates, define I as the set of instances in the current node, and I_L and I_R as the sets of instances in the left and right nodes after the split. The formula for calculating

Algorithm 2 Proposed FCE-XGBoost NLOS/LOS Classification Algorithm**Parameters:**

dtrain: Training data in DMatrix format;

param: XGBoost hyperparameters;

k: Number of folds for cross-validation;

early_stopping_rounds: Early stopping parameter to avoid overfitting;

num_round: Number of boosting rounds.

Input: UWB features from CIR data with FCE scores.

Output: NLOS/LOS classification results.

```

1: Step 1: Convert data to DMatrix format.
2:  $dtrain = \text{DMatrix}(\text{train\_data}, \text{label} = \text{train\_labels})$ 
3: Step 2: Set XGBoost parameters.
    $\text{param} = \{\text{max\_depth} : 6, \text{eta} : 0.05, \text{subsample} : 0.9, \text{colsample\_bytree} : 0.9, \text{objective} : \text{binary}\}$ 
4: Step 3: Perform K-fold Cross-Validation.
    $\text{cv\_results} = \text{xgb.cv}(\text{param}, dtrain, \text{num\_round}, \text{nfold} = 5, \text{metrics} = ['auc'], \text{early\_stopping\_rounds} = 10)$ 
5: Determine best  $\text{num\_round}$  and tune  $\text{param}$  if necessary.
6: Step 4: Train the XGBoost model with optimal parameters.
    $\text{bst} = \text{xgboost.train}(\text{param}, dtrain, \text{num\_round})$ 
7: Step 5: Predict labels on test data.
8:  $\text{predictions} = \text{bst.predict}(dtest)$ 
9: Convert  $\text{predictions}$  to binary labels (0 for LOS, 1 for NLOS)
10: Step 6: Evaluate the model.
11: Calculate accuracy, precision, recall, and F1 score using the test labels.
12: Output: NLOS/LOS classification results, evaluation metrics, and cross-validation results.

```

the reduction in loss is:

$$L_{\text{split}} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma \quad (22)$$

2) XGBoost MODEL ESTABLISHMENT

- The model is initialized by setting the predictions $\hat{y}_i^{(0)}$, usually to a constant value, such as the mean of the target variable.
- The pseudo-residuals $r_i^{(t)}$ are then calculated for each iteration t using the formula:

$$r_i^{(t)} = -\frac{\partial L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}. \quad (23)$$

- A regression tree $f_t(x)$ is constructed using the current pseudo-residuals $r_i^{(t)}$ as the target variable to build a new regression tree $f_t(x)$.
- The model $\hat{y}_i^{(t)}$ is updated by adding the prediction of the new tree to the model $\hat{y}_i^{(t-1)} + \eta f_t(x_i)$, where η is the learning rate, controlling the contribution of each tree to the final prediction.

Algorithm 2 summarizes the FCE-XGBoost LOS/NLOS identification process. Finally, the FCE results were combined with the original features and converted into the

DMatrix format required by XGBoost. The model was then trained using this combined dataset, and its predictions were evaluated by calculating accuracy, precision, recall, and F1 score. These metrics are essential for verifying the model's performance and ensuring that it meets the required standards for LOS/NLOS identification.

IV. EVALUATION AND ANALYSIS

To evaluate the proposed NLOS identification method, this study utilized NLOS/LOS datasets from multiple publicly available benchmarks. These datasets were collected from seven distinct indoor environments, including office, apartment, kitchen, and bedroom, ensuring data diversity and generalizability. In each scenario, we gathered 3,000 NLOS samples and 3,000 LOS samples, resulting in a total of 42,000 samples. The 6,000 samples from each scene were randomly shuffled and then split into a training set comprising 70% of the data (29,400 samples) and a test set comprising 30% of the data (12,600 samples). This strategy of sample selection and partitioning not only prevents the model from overfitting to specific locations but also enhances its generalization capability across different environments.

Each CIR signal in the dataset is represented as a time series vector with 1,016 samples. Detailed information about the dataset is available in [41], and the dataset itself can be accessed on GitHub. By utilizing a diverse array of NLOS/LOS measurement data from various environments, this study effectively verifies the robustness and applicability of the proposed method.

The accuracy, precision, recall, and F1 score of the proposed algorithm were verified through Matlab simulations. Additionally, the performance of the proposed localization algorithm was compared with four existing NLOS identification algorithms using this dataset: MWT-CNN, DT, CNN-LSTM, WGCNN, and FCN-Attention. This comparison further underscores the effectiveness of the proposed method in real-world applications.

A. PERFORMANCE EVALUATION METRICS

In UWB NLOS identification, the evaluation metrics include four aspects: accuracy, precision, recall, and F1 score. These metrics are crucial for assessing the performance of the identification models in various conditions, particularly in distinguishing between LOS and NLOS scenarios.

$$\begin{aligned}
 \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\
 \text{Precision} &= \frac{TP}{TP + FP} \\
 \text{Recall} &= \frac{TP}{TP + FN} \\
 F1 &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (24)
 \end{aligned}$$

where the abbreviations TP, TN, FP, and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively [42]. TP indicates the number of

correctly identified LOS instances, TN represents correctly identified NLOS instances, FP corresponds to LOS instances mistakenly identified as NLOS, and FN represents NLOS instances mistakenly identified as LOS. Precision refers to the proportion of true positive predictions among all positive predictions, while Recall indicates the proportion of true positive predictions among all actual positive instances. The F1 score is the harmonic mean of Precision and Recall, offering a balanced metric that accounts for both precision and recall. Accuracy measures the overall performance of the model in correctly identifying both LOS and NLOS conditions.

In summary, these metrics collectively provide a comprehensive evaluation of the UWB NLOS identification model's performance, enabling researchers to understand its strengths and areas for improvement.

B. FCE-BASED IDENTIFICATION RESULTS

In the initial process of NLOS identification, a FCE method was employed. The implementation began with the definition and calculation of multiple membership functions, including Gaussian and triangular membership functions. These functions were essential for evaluating the data distribution of each feature within the training and testing sets. Specifically, statistical metrics such as the mean and standard deviation for each feature were calculated to define these membership functions. The goal behind selecting and computing these membership functions was to better characterize the behavior patterns of different features under various conditions, thereby enhancing the model's identification capability.

Once the membership functions were established, a regularized objective function was introduced for weight optimization. The regularization terms, including L2 regularization and L1 regularization, were incorporated to prevent model overfitting. Following this, the objective function was defined, and an optimization algorithm, specifically PSa, was employed to optimize the weights. The PSa has demonstrated effectiveness in identifying the global optimal solution. These optimized weights were subsequently used to calculate the FCE scores, which represent the probability of each sample being classified as NLOS.

To further refine identification accuracy, a finer threshold optimization was conducted. The optimal threshold was determined by searching within the range of 0 to 1, using a step size of 0.0001, with the objective of maximizing the F1 score. This process resulted in an optimal threshold of 0.467. The preliminary identification results based on the FCE method were as follows: an accuracy of 0.8524, a precision of 0.8043, a recall of 0.8292, and an F1 score of 0.7992.

Although these results demonstrate a certain level of identification capability, there remains room for improvement. Therefore, future work will focus on integrating the XGBoost algorithm for further optimization. This integration aims to enhance the model's performance, particularly in terms of improving identification accuracy and robustness.

C. FCE-XGBoost RESULTS

The method combining FCE and XGBoost for NLOS identification involves multiple steps to enhance the model's accuracy and robustness. Initially, the dataset is split into training and testing sets using the holdout method, ensuring 70% of the data is used for training and 30% for testing. Subsequently, feature normalization is performed to standardize each feature column, ensuring consistent scaling across all features. To further verify the reliability and generalizability of the model, we employed 5-fold cross-validation. This approach helps in checking whether the generated model is overly dependent or independent of the dataset, ensuring that the model's performance is consistently validated across different data folds before applying it to test data or real-world applications.

During the training phase, Python's XGBoost and NumPy libraries are utilized. The training and testing features are converted into NumPy arrays and subsequently transformed into `DMatrix` objects, an optimized data structure for XGBoost. Key XGBoost parameters are set, including `max_depth` for controlling tree depth, `eta` for learning rate, `subsample` and `colsample_bytree` for managing the proportion of training data and features, and `objective` and `eval_metric` for defining the learning objective and evaluation metrics. Early stopping is employed to train the XGBoost model to prevent overfitting, combining training and evaluation data during the process.

Predictions are made on the test set, and the model's performance is evaluated using a confusion matrix to derive key metrics: accuracy, precision, recall, and F1 score. The results demonstrate superior performance, with an accuracy of 93.13%, a precision of 95.36%, a recall of 90.52%, and an F1 score of 92.88%. This integrated FCE-XGBoost method leverages the strengths of both approaches: FCE for preliminary evaluation and XGBoost for further optimization by learning complex patterns and interactions in the data. The comprehensive process results in a robust model capable of effectively distinguishing between NLOS and LOS conditions, as reflected in the superior evaluation metrics.

To verify the robustness and generalization capability of the developed FCE-XGBoost algorithm in different environments, it is necessary to apply it to various scenarios. Different scenarios have distinct signal propagation characteristics and obstacle distributions, which may affect the identification of NLOS and LOS conditions. By testing the algorithm in multiple real-world scenarios, its performance across different environments can be assessed, ensuring its adaptability and stability.

This approach not only validates the model's robustness but also enhances its generalization ability, ensuring efficient and accurate performance in diverse practical applications. The hyperparameters were applied to seven different scenarios, including Office1, Office2, Small Apartment, Small Workshop, Kitchen Room, Bedroom, and Boiler Room, to test their effectiveness across varying environments. The results show that the chosen hyperparameters have

TABLE 2. Performance metrics across different indoor scenarios.

Scenario	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Office1	93.2	95.4	90.6	92.9
Office2	92.5	95.3	89.2	92.2
Small apartment	93.1	95.8	89.9	92.8
Small workshop	92.1	93.8	90.2	92.0
Kitchen room	92.0	93.7	90.0	91.6
Bedroom	93.7	94.6	92.3	93.4
Boiler room	92.4	94.9	90.0	92.3

minimal impact on performance variations between different scenarios, further demonstrating the stability and robustness of the model under diverse conditions. The remaining six scenarios and the test results are presented in Table 2.

The results presented in the table demonstrate that while the performance of FCE-XGBoost varies across different scenarios, the overall accuracy, precision, recall, and F1 scores remain consistently high, with particularly strong performance in office and bedroom environments. This can be attributed to several factors. First, the use of FCE allows for flexible feature extraction and selection, dynamically adjusting feature weights to adapt to varying environments, thus maintaining high classification accuracy in diverse indoor settings. Second, XGBoost, as a gradient boosting algorithm, excels at handling complex, nonlinear data. Even in more challenging scenarios, such as the boiler room with significant signal interference, the method sustains a high accuracy of 92.4%, demonstrating its robustness in such conditions. Additionally, the F1 scores across all scenarios exceed 91%, indicating the model's effectiveness in handling imbalanced data and adapting to the unique signal reflections and obstructions present in each environment. While FCE-XGBoost performs well overall, performance discrepancies between scenarios are still evident. For instance, the more open office environment, with fewer signal interferences, outperforms more complex environments like the boiler room, where dense equipment increases signal interference.

TABLE 3. Comparative analysis of accuracy metrics for different methods.

Method	Accuracy Avg. $\pm$ Std.(%)	Max.(%)	Min.(%)
CNN-LSTM [27]	78.37 $\pm$ 1.34	81.56	75.60
DT [43]	89.2	—	—
MWT-CNN [28]	80.57 $\pm$ 1.03	81.80	77.90
WGCNN [30]	82.63 $\pm$ 0.89	84.00	81.10
FCN-Attention [29]	87.63 $\pm$ 0.89	88.24	86.10
FCE(proposed)	81.57 $\pm$ 1.28	83.00	80.20
XGboost(proposed)	87.57 $\pm$ 1.03	88.80	86.90
FCE-XGboost(proposed)	93.12 $\pm$ 0.72	93.80	92.10

The test results in these six different scenarios comprehensively validate the adaptability and robustness of the FCE-XGBoost model. The model consistently maintains

TABLE 4. Computation time and performance metrics of various algorithms on the server.

Algorithm	Model Parameters	Calculation Time	Total Evaluated Samples	Calculation Time Per Sample (ms)
CNN-LSTM	1,058,882	7288.984	21,000	0.3471
Decision Tree (DT)	12,345	274.057	21,000	0.0131
MWT-CNN	875,642	5610.945	21,000	0.2672
WGCNN	456,789	2397.042	21,000	0.1141
FCN-Attention	1,097,587	325.871	21,000	0.0155
FCE-XGBoost	44,200	39.704	21,000	0.0019

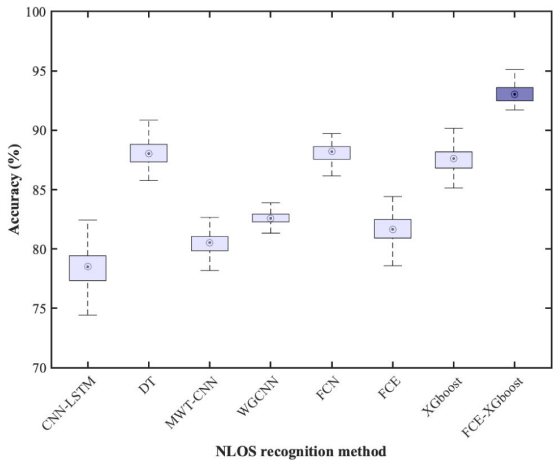


FIGURE 8. Comparison of different NLOS classification models.

high identification performance across various environments, demonstrating good generalization ability and providing stable NLOS and LOS identification results in a wide range of practical application scenarios. This provides strong support for applying the model in extensive indoor positioning and signal transmission contexts. To further validate the effectiveness of the proposed NLOS recognition method, several representative machine learning and deep learning methods were employed for performance comparison. All eight models were trained twenty times, respectively, to avoid the possible random influence, the identification results are shown in in Table 3 and Fig. 8.

In Table 3, “Avg.” represents the average accuracy value, “Std.” represents the standard deviation of the accuracy, “Max.” is the maximum accuracy value, and “Min.” is the minimum accuracy value from the 20 modeling experiments.

It is noteworthy that the LSTM model does not seem to benefit classification performance. With the use of LSTM, the average accuracy of the CNN-LSTM method [27] even decreases to 78.37%. By employing wavelet analysis, the performance of MWT-CNN [28] is slightly higher, with an accuracy of 80.57%. With the introduction of GAF, WGCNN [30] achieves an average accuracy of 82.63%. This indicates that the contribution of GAF to performance improvement is greater than that of wavelet analysis.

The FCN-Attention algorithm leverages the deep feature extraction capabilities of a convolutional neural network based on FCN and the information enhancement abilities of the self-attention mechanism, achieving a better classification accuracy of 88.24%. The classification performance of the DT method [43] is also quite promising, achieving an accuracy of 89.2%, surpassing that of other comparative classification algorithms.

D. PERFORMANCE EVALUATION AND COMPARISON OF THE FCE-XGBoost ALGORITHM

To comprehensively evaluate the performance of each algorithm, this study analyzes both the total number of parameters and the runtime on each dataset. The experiments were conducted on a Lenovo ThinkSystem SR630 server, equipped with an Intel Xeon Silver 4210 CPU (10 cores, 2.2 GHz), NVIDIA RTX 2080 Ti GPU, and 16 GB DDR4 RAM. For comparison, additional experiments were carried out on a Lenovo ThinkStation P340 workstation, featuring an Intel Xeon E3-1240 v6 CPU (4 cores, 3.7 GHz), NVIDIA GTX 1060 GPU, and 16 GB DDR4 RAM. To ensure fairness, both the server and workstation used the same datasets, and the computation time was measured solely based on the classification process. The results, as shown in Table 4 and Table 5, illustrate the computational efficiency of each algorithm on the two hardware platforms.

Upon comparing the performance of different algorithms across both platforms, our proposed FCE-XGBoost algorithm demonstrated superior results, particularly in terms of computational efficiency, significantly outperforming the other algorithms. On the Dell T630 server, the FCE-XGBoost algorithm achieved a per-sample computation time of only 0.0019 ms, while on the Dell T3620 workstation, it required only 0.0022 ms per sample, making it the fastest among all evaluated algorithms. This indicates that FCE-XGBoost exhibits excellent computational efficiency and real-time capability for handling NLOS/LOS classification tasks. In contrast, while CNN-LSTM and FCN-Attention have larger model sizes, with 1,058,882 and 1,097,587 parameters respectively, their computation times are relatively longer. Notably, CNN-LSTM took 0.3471 ms and 0.1893 ms per sample on the server and workstation, respectively. By contrast, FCE-XGBoost not only maintained a smaller model size (with only 44,200 parameters) but also significantly reduced

TABLE 5. Computation time and performance metrics of various algorithms on the workstation.

Algorithm	Model Parameters	Calculation Time	Total Evaluated Samples	Calculation Time Per Sample (ms)
CNN-LSTM	1,058,882	3974.485	21,000	0.1893
Decision Tree (DT)	12,345	300.000	21,000	0.0143
MWT-CNN	875,642	6064.352	21,000	0.2888
WGCNN	456,789	2654.586	21,000	0.1264
FCN-Attention	1,097,587	356.734	21,000	0.0170
FCE-XGBoost	44,200	45.360	21,000	0.0022

computation time, thereby greatly enhancing the real-time processing capabilities. Therefore, the FCE-XGBoost algorithm stands out in terms of both model size and computational efficiency, making it particularly suitable for real-time applications, and presenting substantial potential for practical use.

Furthermore, the proposed FCE method also shows significant performance improvement, with an average accuracy of 81.57%. However, the most notable improvements are observed with the XGBoost and FCE-XGBoost methods, where the average accuracies are 87.57% and 93.12%, respectively. This indicates that combining FCE with XGBoost can significantly enhance the accuracy of NLOS recognition.

V. CONCLUSION

In this study, we propose a novel UWB NLOS/LOS classification method based on the FCE-XGBoost algorithm. The approach begins with selecting six UWB CIR feature data points using PCC to minimize multicollinearity. These features are then standardized to ensure uniformity in the input data. Subsequently, the membership degrees for each feature are calculated, and the comprehensive membership degree is derived by combining these individual degrees.

To further enhance classification performance, the PSa is employed to adjust feature weights, optimizing the FCE score. Through threshold optimization, we determine the optimal threshold that yields the highest classification accuracy. The optimized FCE scores are then used as input features for training the XGBoost model. The trained model is subsequently evaluated using a test dataset to assess its classification performance.

The results indicate that the FCE-XGBoost method achieves an average classification accuracy of 92.57% across multiple simulations, outperforming several existing algorithms applied to the same dataset [26], [27]. Moreover, the system demonstrates stable performance across key metrics, including accuracy, precision, recall, and F1 score, underscoring the robustness and generalization capability of the proposed model. This consistent performance suggests that the FCE-XGBoost algorithm is highly effective for UWB NLOS/LOS classification and can be reliably deployed in different environments. Additionally, we evaluated the model's performance on both a server and a workstation,

confirming its computational efficiency across different hardware platforms.

Despite the promising results, it is important to acknowledge that this study did not investigate the influence of NLOS/LOS classification accuracy on subsequent location estimation. Future work should explore integrating NLOS/LOS classification accuracy into location estimation models, as this could further improve system performance in real-world applications. Furthermore, we did not assess the impact of diffraction effects in UWB-based indoor positioning systems due to limitations in both the hardware and the dataset. Investigating these diffraction effects could provide deeper insights into signal propagation behavior in complex environments and represents a critical avenue for future research.

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